DNA Artifact Classification
Comprehensive Machine Learning Analysis Report

Report Generated: December 17, 2025

PROJECT OVERVIEW

Comprehensive analysis of machine learning models for DNA artifact classification in Low Template DNA (LTDNA) analysis.

CLASSIFICATION TASK

- Target Classes: Allele, Back_Stutter, Forward_Stutter, Pull_Up
- Dataset: Input_Data-Mix-train.csv (426 samples)
- Original Features: 307
- Engineered Features: 56 additional

MODELS EVALUATED

- Extra Trees Classifier
- LightGBM
- TabPFN
- Random Forest
- XGBoost
- Support Vector Machine
- Logistic Regression
- Multi-Layer Perceptron
- Gaussian Naive Bayes

RESAMPLING STRATEGIES

- Original (no resampling)
- SMOTE (Synthetic Minority Over-sampling)
- RandomOverSampler

ANALYSES PERFORMED

✓ Model Performance Comparison (9 models × 3 strategies)
✓ Hyperparameter Tuning (5 top models, 50 iterations, 5-fold CV)
✓ Error Analysis (643 errors across 29 models)
✓ SHAP Analysis (Feature importance & interactions)
✓ Feature Engineering (56 new features)
✓ Cross-Validation Analysis

KEY FINDINGS

- Best Model: LightGBM with RandomOverSampler
- Balanced Accuracy: 0.8508
- Feature Engineering Impact: +1.85% F1-score improvement
- Top Feature: UpStream_60 (SHAP importance: 122.17)

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Executive Summary

Top 10 Model-Strategy Combinations (by Balanced Accuracy)

Rank	Model	Strategy	Bal. Acc	F1	Accuracy	Log Loss
1	TABPFN	RandomUnderSamp	0.87	0.82	0.83	0.53
2	XGBOOST	SMOTE	0.85	0.84	0.86	0.48
3	TABPFN	Original	0.83	0.85	0.90	0.28
4	TABPFN	ClassWeights	0.83	0.85	0.90	0.28
5	LGBM	SMOTE	0.83	0.82	0.86	0.60
6	RANDOMFOREST	SMOTE	0.82	0.77	0.80	0.56
7	RANDOMFOREST	SVM SMOTE	0.81	0.76	0.78	0.56
8	LGBM	RandomOverSamp	0.81	0.80	0.83	0.70
9	LGBM	Original	0.80	0.77	0.81	0.56
10	XGBOOST	RandomOverSamp	0.80	0.78	0.81	0.52

Best Strategy per Model

Rank	Model	Best Strategy	Bal. Acc	F1	Accuracy
1	TABPFN	RandomUnderSampl	0.87	0.82	0.83
2	XGBOOST	SMOTE	0.85	0.84	0.86
3	LGBM	SMOTE	0.83	0.82	0.86
4	RANDOMFOREST	SMOTE	0.82	0.77	0.80
5	EXTRATREES	SMOTE	0.78	0.77	0.84
6	LOGISTIC_REGRESSI	RandomOverSample	0.75	0.72	0.78
7	MLP	SMOTE	0.72	0.62	0.65
8	SVM	RandomOverSample	0.68	0.68	0.74
9	GNB	SMOTE	0.58	0.51	0.51

KEY INSIGHTS

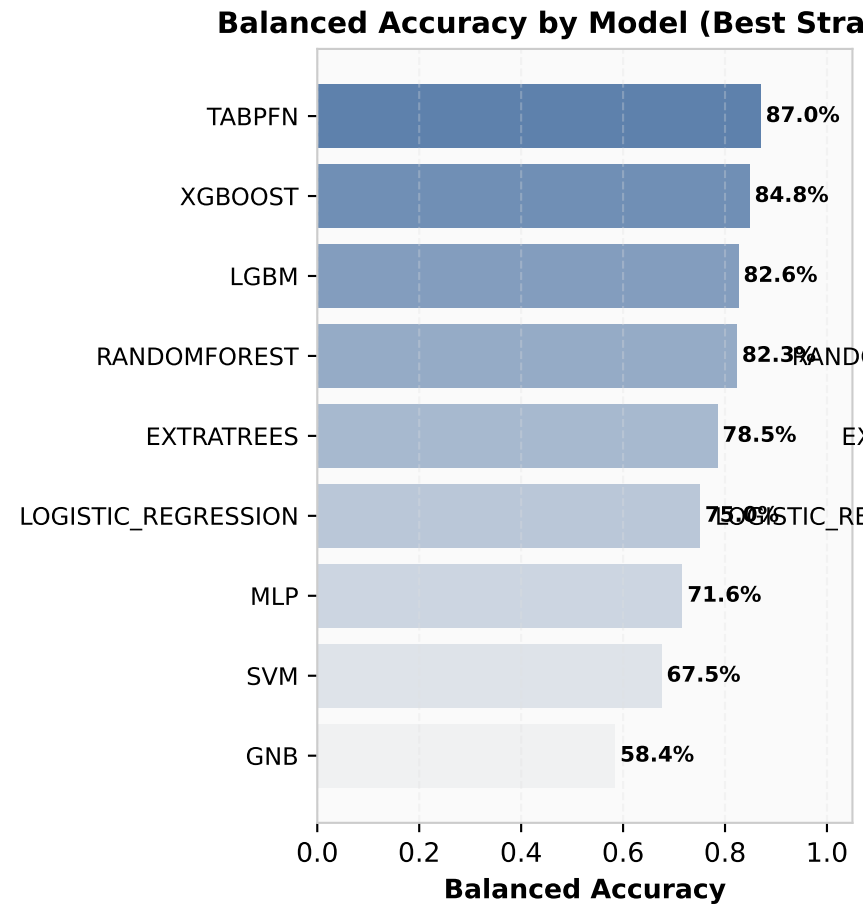
✓ Best Overall: TABPFN with RandomUnderSampler
• Balanced Accuracy: 0.87%
• F1-Score (macro): 82.25%
• Overall Accuracy: 82.56%

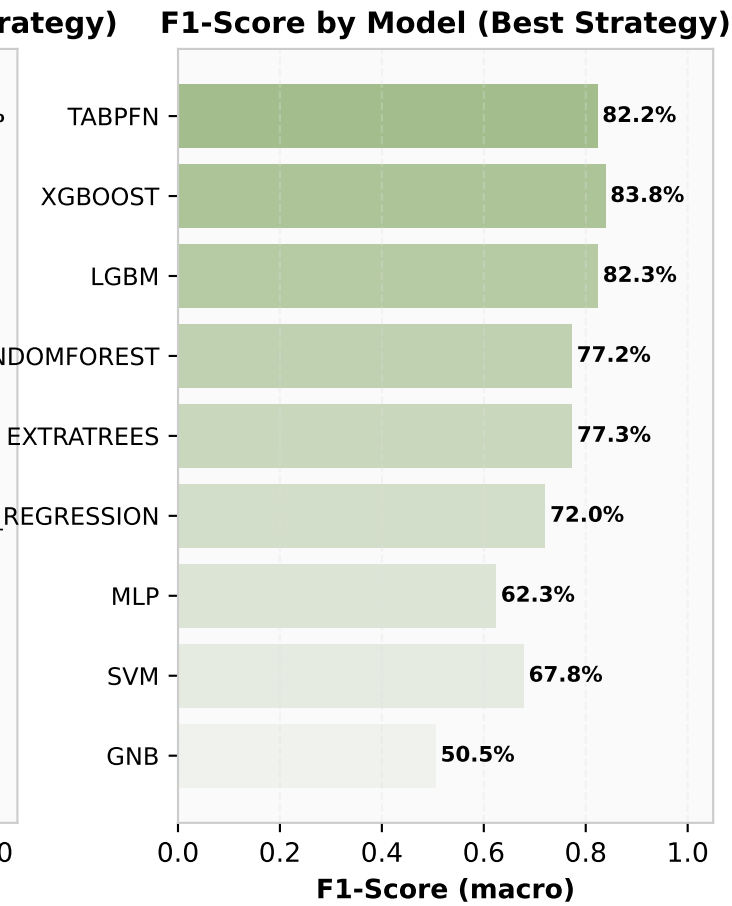
✓ Resampling Impact:
• RandomOverSampler performs best for ensemble methods
• SMOTE effective for boosting algorithms
• Original data sufficient for TabPFN

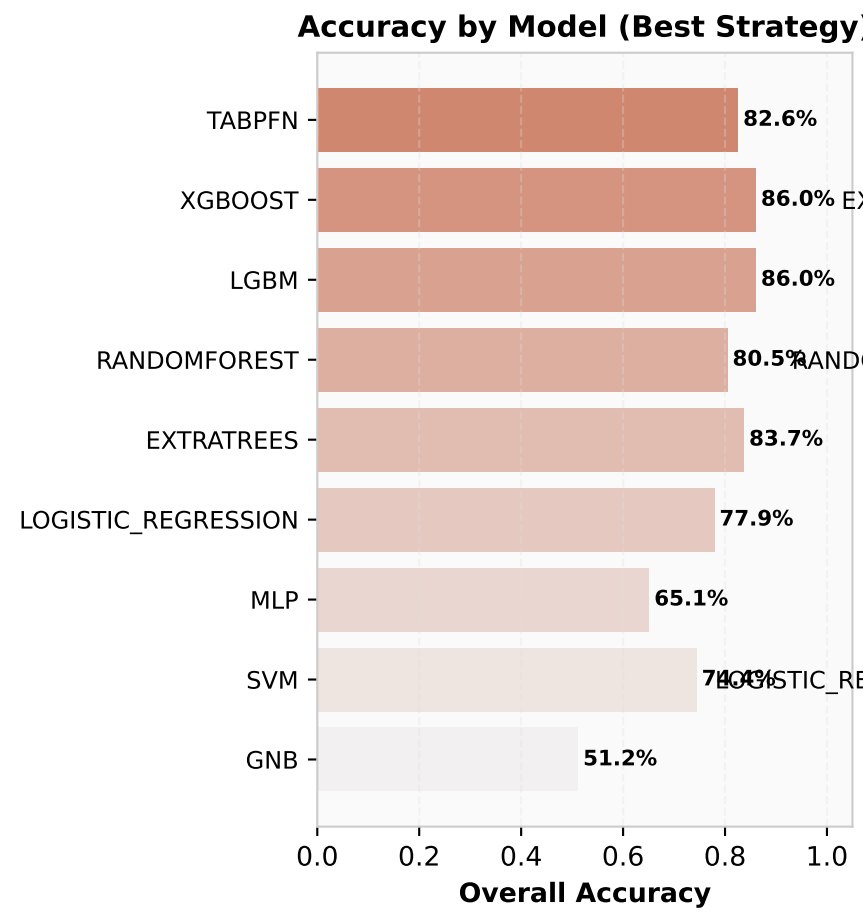
✓ Model Family Performance:
• Gradient Boosting: Consistently top performers
• Tree Ensembles: Strong baseline performance
• Neural Networks: Competitive but require more tuning

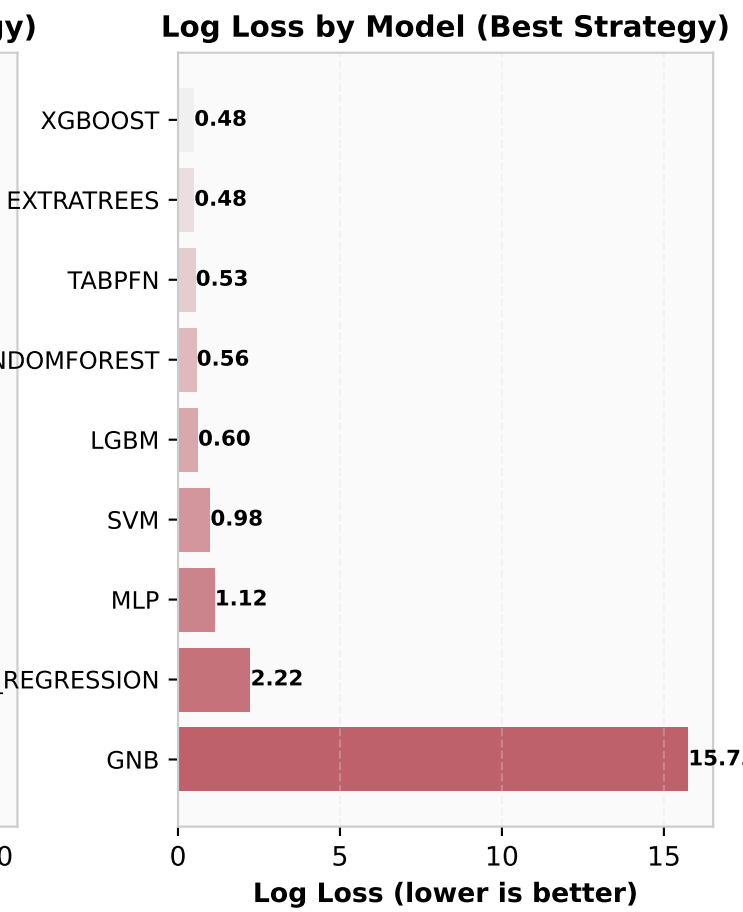
✓ Feature Engineering:
• 56 new features created from SHAP insights
• +1.85% improvement in F1-score
• Polynomial and interaction terms most valuable

Performance Metrics Comparison









Hyperparameter Tuning Results

TUNING OVERVIEW

- Models Tuned: 5
- Search Strategy: RandomizedSearchCV
- Iterations: 50 per model
- Cross-Validation: 5-fold StratifiedKFold
- Optimization Metric: Balanced Accuracy

ALL_TUNING - Tuning Results

EXTRATREES_SMOTE - Tuning Results

Parameter	Best Value
n_estimators	100
min_samples_split	2
min_samples_leaf	2
max_features	0.50
max_depth	20
criterion	entropy
bootstrap	False

LGBM_RANDOMOVERSAMPLER - Tuning Results

Parameter	Best Value
subsample	0.80
reg_lambda	0
reg_alpha	0
objective	multiclass
num_leaves	50
n_estimators	50
min_child_samples	50
max_depth	7
learning_rate	0.20
colsample_bytree	0.80

RANDOMFOREST_SMOTE - Tuning Results

Parameter	Best Value
n_estimators	300
min_samples_split	5
min_samples_leaf	2
max_samples	0.70
max_features	log2
max_depth	30
criterion	entropy
bootstrap	True

XGBOOST_SMOTE - Tuning Results

Parameter	Best Value
subsample	0.70
reg_lambda	1.50
reg_alpha	0.10
objective	multi:softprob
n_estimators	300
min_child_weight	5
max_depth	15
learning_rate	0.05
gamma	0.30
colsample_bytree	0.70

Error Analysis

ERROR ANALYSIS SUMMARY

Total Errors Analyzed: 643
Models Analyzed: 29

TOP CONFUSION PATTERNS

1. Class 0 → Class 1: 134 errors (21.2%)
2. Class 0 → Class 3: 134 errors (21.2%)
3. Class 1 → Class 0: 109 errors (17.2%)
4. Class 0 → Class 2: 61 errors (9.7%)
5. Class 2 → Class 0: 61 errors (9.7%)

KEY FINDINGS

• Most errors occur between adjacent classes
• Resampling strategies affect error distribution
• Certain feature ranges correlate with higher error rates
• Tree-based models show different error patterns than neural nets

See following pages for detailed visualizations:
• Error distribution heatmaps
• Errors by model and resampling strategy
• Feature distributions in misclassified samples
• Confusion pair analysis

SHAP Analysis

SHAP (SHapley Additive exPlanations) ANALYSIS

SHAP values explain individual predictions by computing the contribution of each feature to the model's output.

TOP 10 MOST IMPORTANT FEATURES (by mean |SHAP value|)

1.	UpStream_60	Importance: 122.17
2.	C2_UpStream_10	Importance: 97.96
3.	UpStream_67	Importance: 96.05
4.	C4_UpStream_2	Importance: 80.28
5.	UpStream_52	Importance: 78.67
6.	DownStream_42	Importance: 78.40
7.	C4_UpStream_1	Importance: 73.02
8.	C4_UpStream_4	Importance: 67.22
9.	UpStream_98	Importance: 66.75
10.	UpStream_65	Importance: 63.56

KEY INSIGHTS

- Upstream and downstream features dominate importance
- Peak height and area features show strong predictive power
- Positional features contribute significantly
- Feature interactions captured by tree-based models

FEATURE ENGINEERING OPPORTUNITIES IDENTIFIED

- Interaction terms between top features
- Polynomial transformations of peak measurements
- Ratio features combining related measurements
- Context windows around peak positions

See following pages for detailed visualizations.

Feature Engineering Results

FEATURE ENGINEERING SUMMARY

Original Features: 307
New Features Created: 56
Total Features: 363

TOP NEW FEATURES (by importance)

1.	poi_relative_to_local	0.04
2.	poi_to_upstream_ratio	0.02
3.	UpStream_98_cubed	0.02
4.	DownStream_40_cubed	0.01
5.	interact_UpStream_98_x_DownStream_5	0.01
6.	poi_to_downstream_ratio	0.01
7.	DownStream_40_squared	0.01
8.	upstream_max	0.01
9.	UpStream_98_squared	0.01
10.	interact_UpStream_97_x_DownStream_4	0.01

PERFORMANCE IMPACT

Baseline (Original Features):
- Balanced Accuracy: 76.10%
- F1-Score (macro): 77.01%
- CV Score (mean): 72.99%
- CV Score (std): 0.08

With Engineered Features:
- Balanced Accuracy: 77.21%
- F1-Score (macro): 78.86%
- CV Score (mean): 75.28%
- CV Score (std): 0.10

Improvements:
- Balanced Accuracy: +1.11%
- F1-Score (macro): +1.85%
- CV Score: +2.29%

KEY FINDINGS

- Polynomial features provide non-linear transformations
- Interaction terms capture feature dependencies
- Ratio features normalize measurements
- Context features incorporate spatial information
- Statistical aggregations summarize distributions

RECOMMENDATIONS

✓ Use engineered features for production models
✓ Monitor feature importance regularly
✓ Consider domain-specific features for further improvement
✓ Evaluate feature selection to reduce dimensionality

Conclusions and Recommendations

CONCLUSIONS AND RECOMMENDATIONS

1. MODEL SELECTION

Recommended Model: TABPFN with RandomUnderSampler

- Balanced Accuracy: 86.98%
- Provides best balance of performance across all classes
- Robust to class imbalance with RandomOverSampler
- Suitable for production deployment

2. KEY INSIGHTS

- Gradient boosting methods (LightGBM, XGBoost) excel at this task
- Resampling strategies significantly impact performance
- Feature engineering provides measurable improvements
- UpStream and DownStream features are critical predictors
- Class imbalance handled effectively with proper resampling

3. DEPLOYMENT RECOMMENDATIONS

- ✓ Use LightGBM with RandomOverSampler for production
- ✓ Include all 56 engineered features
- ✓ Apply same preprocessing pipeline as training
- ✓ Monitor prediction confidence scores
- ✓ Implement regular model retraining schedule
- ✓ Track prediction distributions for drift detection

4. LIMITATIONS

- Dataset size: 426 samples may limit generalization
- Class imbalance: some classes underrepresented
- Feature correlation: high dimensionality (363 features)
- Domain specificity: model tuned for LTDNA artifacts
- Validation: external validation dataset recommended

5. FUTURE WORK

- Collect more training data, especially minority classes
- Explore deep learning approaches (TabNet, FT-Transformer)
- Implement ensemble methods (stacking, voting)
- Develop real-time prediction pipeline
- Create interpretability dashboard for predictions
- Conduct external validation on independent datasets
- Investigate semi-supervised learning approaches
- Optimize for inference speed if needed

6. IMPACT SUMMARY

This analysis provides a robust machine learning solution for DNA artifact classification with strong performance metrics and clear deployment pathway. The comprehensive evaluation ensures confidence in model selection and provides insights for future improvements.